Lecture 12 recap

In Lecture 12:

- Learnt how the LT maps derivative in t-space to multiplication in s-space
- Understood how the LT encodes initial values
- Learnt how to solve IVPs using the LT and ILT
- Introduced the Heaviside step function
- Learnt how to use the s- and t-shifting theorems to calculate LTs and ILTs

Lecture 13

In Lecture 13, we will:

1. Learn how to solve IVPs involving a forcing that "switches on"
2. Introduce the Dirac delta

Discontinuous IVPs; **Exercise 3.17**: $y'(t) + y(t) = H(t-2),\ y(0) = 1$

$H(t-2)$ on right-hand side $\Rightarrow$ forcing function switches on at $t = 2$.

$$\underbrace{\mathscr{L}\{y'(t) + y(t)\}}_{sY - 1 + Y} = \underbrace{\mathscr{L}\{H(t-2)\}}_{\mathrm{e}^{-2s}\mathscr{L}(1)} = \frac{\mathrm{e}^{-2s}}{s} \qquad t\text{-shifting}$$

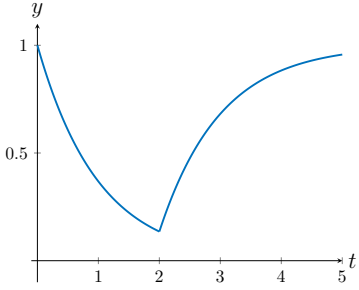
$$\Rightarrow \quad Y = \frac{1}{s+1} + \frac{\mathrm{e}^{-2s}}{s(s+1)}$$

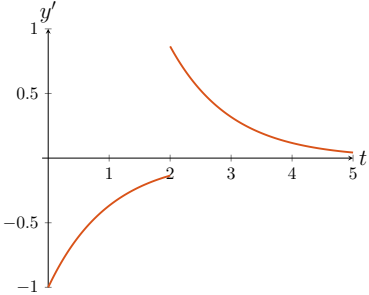
$$\frac{1}{s(s+1)} = \frac{A}{s} + \frac{B}{s+1} \ \Rightarrow\ A(s+1) + Bs = 1 \ \Rightarrow\ A = 1, B = -1$$

$$\Rightarrow \quad Y = \frac{1}{s+1} + \frac{\mathrm{e}^{-2s}}{s} - \frac{\mathrm{e}^{-2s}}{s+1}$$

$$y = \underbrace{\mathscr{L}^{-1}\left\{\frac{1}{s+1}\right\}}_{\mathrm{e}^{-t}} + \underbrace{\mathscr{L}^{-1}\left\{\frac{\mathrm{e}^{-2s}}{s}\right\}}_{H(t-2)} - \underbrace{\mathscr{L}^{-1}\left\{\frac{\mathrm{e}^{-2s}}{s+1}\right\}}_{\mathrm{e}^{-(t-2)}H(t-2)} = \mathrm{e}^{-t} + (1-\mathrm{e}^{2-t})H(t-2)$$

$$\lim_{t\to 2^-} y(t) = \lim_{t\to 2^+} y(t) = \mathrm{e}^{-2} \quad \text{but} \quad \lim_{t\to 2^-} y'(t) = -\mathrm{e}^{-2} \neq 1-\mathrm{e}^{-2} = \lim_{t\to 2^+} y'(t)$$





Solution to IVPs with discontinuous forcing

Consider the nth-order ODE

$$y^{(n)}(t) + f_{n-1}(t)y^{(n-1)} + \ldots + f_0(t)y(t) = g(t)$$

- Let $f_0, \ldots, f_n$ be continuous on $[0, \infty)$
- Let g be piecewise continuous on $[0, L]$ for all $L > 0$

ODE has solutions, such that:

1. $y, y', \ldots, y^{(n-1)}$ are continuous on $[0, \infty)$
2. $y^{(n)}$ is continuous everywhere g is
3. The left and right limits of $y^{(n)}$ are defined by the ODE at all discontinuities
4. y satisfies the ODE at all points of continuity of g

Exercise 3.18: Solve the below IVP

$$y'' + 4y = g(t) \quad \text{where} \quad g(t) = \begin{cases} t & 0 < t \leqslant 2 \\ 0 & t > 2 \end{cases} \quad \text{and} \quad y(0) = y'(0) = 0$$

Rewrite the discontinuous term

$$g(t) = t - t\,H(t-2) = t - 2\,H(t-2) - (t-2)\,H(t-2)$$

Taking the Laplace transform of both sides:

$$\mathscr{L}\{y'' + 4\,y\} = s^2 Y + 4Y = \underbrace{\mathscr{L}\{t\}}_{\frac{1}{s^2}} - 2\,\underbrace{\mathscr{L}\{H(t-2)\}}_{\frac{\mathrm{e}^{-2s}}{s}} - \underbrace{\mathscr{L}\{(t-2)\,H(t-2)\}}_{\frac{\mathrm{e}^{-2s}}{s^2}}$$

$$\Rightarrow \quad Y(s) = \frac{1}{s^2(s^2+4)} - \frac{2\,s+1}{s^2\,(s^2+4)}\,\mathrm{e}^{-2s}$$

Use partial fractions to write Y in a more convenient form

$$Y = \frac{1}{4s^2} - \frac{1}{4(s^2+4)} - \mathrm{e}^{-2s}\left(\frac{1}{2s} + \frac{1}{4s^2} - \frac{s}{2(s^2+4)} - \frac{1}{4(s^2+4)}\right)$$

$$\Rightarrow \quad y(t) = \mathscr{L}^{-1}\{Y(s)\} = \frac{1}{4}\mathscr{L}^{-1}\left\{\frac{1}{s^2}\right\} - \frac{1}{8}\mathscr{L}^{-1}\left\{\frac{2}{s^2+2^2}\right\} - \frac{1}{2}\mathscr{L}^{-1}\left\{\frac{\mathrm{e}^{-2s}}{s}\right\}$$

$$- \frac{1}{4}\mathscr{L}^{-1}\left\{\frac{\mathrm{e}^{-2s}}{s^2}\right\} + \frac{1}{2}\mathscr{L}^{-1}\left\{\frac{s\,\mathrm{e}^{-2s}}{s^2+4}\right\} + \frac{1}{8}\mathscr{L}^{-1}\left\{\frac{2\,\mathrm{e}^{-2s}}{s^2+4}\right\}$$

$$= \frac{1}{4}t - \frac{1}{8}\sin 2t$$

$$- \left(\frac{1}{2} + \frac{t-2}{4} - \frac{1}{2}\cos 2(t-2) - \frac{1}{8}\sin 2(t-2)\right) H(t-2)$$

$$y(t) = \frac{1}{4}t - \frac{1}{8}\sin 2t - \left(\frac{1}{2} + \frac{t-2}{4} - \frac{1}{2}\cos 2(t-2) - \frac{1}{8}\sin 2(t-2)\right) H(t-2)$$

